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$$1. f(x) = \frac{x}{x-2} = \sum_{n=0}^{\infty} a_n (x-1)^n$$

여기서 $a_7 = \frac{f^{(7)}(1)}{7!}$ 인데 미분을 해서 직접 계산하는 것 보다는 공식을 활용하는 풀이가 더 좋다.

$$\begin{aligned} \frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n \text{이므로} \\ f(x) &= \frac{x}{x-2} = 1 + \frac{2}{x-2} \\ &= 1 + \frac{2}{(x-1)-1} \\ &= 1 - \frac{2}{1-(x-1)} \\ &= 1 - 2 \sum_{n=0}^{\infty} (x-1)^n \end{aligned}$$

따라서 $a_7 = -2$ 이다.

$$2. (x-\pi)^3 \sin x = \sum_{n=0}^{\infty} a_n (x-\pi)^n$$

여기서 $a_6 = \frac{f^{(6)}(\pi)}{6!}$ 인데 미분을 해서 직접 계산하는 것 보다는 공식을 활용하는 풀이가 더 좋다.

$$\begin{aligned} \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \text{이므로} \\ f(x) &= (x-\pi)^3 \sin x \\ &= (x-\pi)^3 \sin(x-\pi+\pi) \\ &= -(x-\pi)^3 \sin(x-\pi) \\ &= -(x-\pi)^3 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (x-\pi)^{2n+1} \end{aligned}$$

따라서 $a_6 = \frac{1}{3!}$ 이다.

$$3. f(x) = \frac{4}{2x+3} = \sum_{n=0}^{\infty} \frac{f^{(n)}(-1)}{n!} (x+1)^n$$

여기서 7차항의 계수는 $\frac{f^{(7)}(-1)}{7!}$ 인데 미분을 해서 직접 계산하는 것 보다는 공식을 활용하는 풀이가 더 좋다.

$$\begin{aligned} \frac{1}{1+x} &= \sum_{n=0}^{\infty} (-1)^n x^n \text{이므로} \\ f(x) &= \frac{4}{2x+3} = \frac{4}{1+2(x+1)} \\ &= 4 \sum_{n=0}^{\infty} (-1)^n (2(x+1))^n \end{aligned}$$

따라서 7차항의 계수는 -4×2^7 이다.

4.

$$\begin{aligned} f'(x) &= \sin(x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (x^2)^{2n+1} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{4n+2} \\ f(x) &= \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!(4n+3)} x^{4n+3} + C \end{aligned}$$

$$a_3 = \frac{1}{3}, a_7 = -\frac{1}{42} \text{이고 } a_3 + a_7 = \frac{13}{42} \text{이다.}$$

$$5. f(x) = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 \text{에서}$$

$$f'(x) = 6x^5 + 5x^4 + 4x^3 + 3x^2 + 2x + 1 \text{이고 } f'(2) = 321 \text{이다.}$$

$$\begin{aligned} \text{또한, } f(2) &= 127 \text{이므로 } x=2 \text{에서 선형근사식은} \\ y &= f(2) + f'(2)(x-2) = 127 + 321(x-2) \text{이므로} \\ f(2.01) &\approx 130.21 \text{이다.} \end{aligned}$$

$$6. f(x) = e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 \dots$$

$$f(0.1) = e^{0.1} = 1 + (0.1) + \frac{1}{2}(0.1)^2 + \frac{1}{6}(0.1)^3 + \dots$$

$$\text{여기서 } \frac{1}{6}(0.1)^3 = 0.00016 \text{이므로 소수점 이하 셋째자리의 수는}$$

$$\frac{1}{6}(0.1)^3 \text{ 앞에 숫자들만 계산해보도 된다.}$$

$$\text{따라서 } e^{0.1} = 1.105 \dots \text{이므로 소수점 이하 셋째자리의 수는 5이다.}$$

7.

$$\begin{aligned} f(x) &= \frac{8}{(2+x)(2-3x)} \\ &= \frac{1}{2+x} + \frac{3}{2-3x} \\ &= \frac{1}{2} \frac{1}{1+\frac{x}{2}} + \frac{1}{2} \frac{3}{1-\frac{3x}{2}} \\ &= \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{x}{2}\right)^n + \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{3x}{2}\right)^n \end{aligned}$$

$$1) \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{x}{2}\right)^n \quad (\text{비율}) \Rightarrow \frac{|x|}{2} < 1 \quad \therefore |x| < 2 = R$$

$$2) \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{3x}{2}\right)^n \quad (\text{비율}) \Rightarrow \frac{3|x|}{2} < 1 \quad \therefore |x| < \frac{2}{3} = R$$

$$\text{둘 다 동시에 수렴해야 } \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{x}{2}\right)^n + \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{3x}{2}\right)^n \text{이 수렴하므로}$$

$$f(x) \text{의 수렴반경은 } \frac{2}{3} \text{이다.}$$

8.

$$f(x) \text{는 기함수 이므로 } a_4 = 0 \text{이다.}$$

$$\begin{aligned} f(x) &= \frac{\sinh x}{\cos x} \\ &= \sinh x \sec x \\ &= \left(x + \frac{1}{3!}x^3 + \frac{1}{5!}x^5 + \dots\right) \left(1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + \dots\right) \end{aligned}$$

$$\text{여기서 } a_3 \text{는 3차항의 계수이므로 구하면 } a_3 = \frac{1}{2} + \frac{1}{6} = \frac{2}{3} \text{이다.}$$

9.

$$\begin{aligned} f(x) &= \frac{\cos x}{1+x^2+x^4} \\ &= \cos x \frac{1}{1+(x^2+x^4)} \\ &= \left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \dots\right) \left(1 - (x^2+x^4) + (x^2+x^4)^2 - (x^2+x^4)^3 + \dots\right) \end{aligned}$$

$p(x)$ 는 5차 테일러 다항식이므로 $p(x) = 1 - \frac{3}{2}x^2 + \frac{13}{24}x^4$ 이고

모든 항의 계수 합을 구하면 $p(1) = 1 - \frac{3}{2} + \frac{13}{24} = \frac{1}{24}$ 이다.

10.

$f(x) = \ln(\cos x)$ 는 우함수이므로 x^3 의 계수는 0이다.

$$\begin{aligned} f(x) &= \ln(\cos x) \\ &= \ln\left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \dots\right) \\ &= \ln\left(1 + \left(-\frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \dots\right)\right) \end{aligned}$$

$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \dots$ 이므로

$$f(x) = \left(-\frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \dots\right) - \frac{1}{2}\left(-\frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \dots\right)^2 + \dots \text{이다.}$$

여기서 x^2 의 계수는 $-\frac{1}{2}$ 이다.

11.

$$\begin{aligned} f(x) &= \ln(1 + \cos x) \\ &= \ln\left(1 + \left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \dots\right)\right) \end{aligned}$$

여기서 $\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \dots$ 에

x 대신 $\left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \dots\right)$ 을 대입하면 1이 같이 들어가서

원하는 차수를 골라내기 어려워진다. 따라서

$$\begin{aligned} &\ln\left(1 + \left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \dots\right)\right) \\ &= \ln\left(2 + \left(-\frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \dots\right)\right) \\ &= \ln\left(2\left(1 + \left(-\frac{1}{4}x^2 + \frac{1}{48}x^4 - \dots\right)\right)\right) \\ &= \ln 2 + \ln\left(1 + \left(-\frac{1}{4}x^2 + \frac{1}{48}x^4 - \dots\right)\right) \\ &= \ln 2 + \left(-\frac{1}{4}x^2 + \frac{1}{48}x^4 - \dots\right) - \frac{1}{2}\left(-\frac{1}{4}x^2 + \frac{1}{48}x^4 - \dots\right)^2 + \dots \text{이고} \end{aligned}$$

$$p(x) = \ln 2 - \frac{1}{4}x^2 - \frac{1}{96}x^4 \text{이다.}$$

즉, $p_4(1) = \ln 2 - \frac{25}{96}$ 이다.

$$12. f(x) = 2x^2\sqrt{1+x^3}$$

$$= 2x^2(1+x^3)^{\frac{1}{2}}$$

$$(1+x)^p = 1 + px + \frac{p(p-1)}{2!}x^2 - \dots \text{이므로}$$

$$\begin{aligned} 2x^2(1+x^3)^{\frac{1}{2}} &= 2x^2\left(1 + \frac{1}{2}x^3 + \dots + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)\left(-\frac{7}{2}\right)}{5!}x^{15} + \dots\right) \\ &= \dots + \frac{1 \times 3 \times 5 \times 7}{2^4 \times 5!}x^{17} + \dots \end{aligned}$$

$$\begin{aligned} \text{여기서 } f^{(17)}(0) &= 17! \times \frac{f^{(17)}(0)}{17!} \\ &= 17! \times \frac{1 \times 3 \times 5 \times 7}{2^4 \times 5!} \\ &= 17! \times \frac{1 \times 2 \times 3 \times 4 \times 5 \times 7}{2^4 \times 5! \times 2 \times 4} \\ &= 17! \times \frac{7}{2^7} \text{이다.} \end{aligned}$$

13.

$$\begin{aligned} f(x) &= \ln(1+x^{10})\tan^{-1}(x^{10}) \\ &= \left(x^{10} - \frac{1}{2}x^{20} + \frac{1}{3}x^{30} - \frac{1}{4}x^{40} - \dots\right)\left(x^{10} - \frac{1}{3}x^{30} - \dots\right) \\ \frac{f^{(40)}(0)}{40!} &= -\frac{1}{3} + \frac{1}{3} = 0 \text{ (40차항의 계수)} \end{aligned}$$

14.

$f(x) = \cos(\arcsin(x^2))$ 을 직접 전개하면 복잡하다.

$\sin^{-1}(x^2) = \alpha$ 라 두면 $\cos \alpha = \sqrt{1-x^4}$ 이다.

$$\text{따라서 } f(x) = (1-x^4)^{\frac{1}{2}} = \dots - \frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}x^{12} + \dots \text{이므로}$$

$$\frac{f^{(12)}(0)}{12!} = -\frac{1}{16} \text{이다.}$$

15.

$$\begin{aligned} &\int_0^1 \sqrt{1+x^3} dx \\ &= \int_0^1 (1+x^3)^{\frac{1}{2}} dx \\ &= \int_0^1 1 + \frac{1}{2}x^3 - \frac{1}{8}x^6 + \frac{1}{16}x^9 - \dots dx \\ &= 1 + \frac{1}{8} - \frac{1}{56} + \frac{1}{160} - \dots \end{aligned}$$

따라서 오차 0.01이내로 근사한 값은 $1 + \frac{1}{8} - \frac{1}{56} = \frac{31}{28}$ 이다.

16.

$$\begin{aligned} &\int_0^1 e^{-x^2} dx \\ &= \int_0^1 1 - x^2 + \frac{1}{2!}x^4 - \frac{1}{3!}x^6 + \dots dx \\ &= 1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \dots \text{이므로} \\ \left| \int_0^1 e^{-x^2} dx - A \right| &< \frac{1}{42} \text{을 만족하는 } A \text{는 } 1 - \frac{1}{3} + \frac{1}{10} = \frac{23}{30} \text{이다.} \end{aligned}$$

17.

$$\begin{aligned} \ln(1+x) &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1} \text{이므로} \\ \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} \left(\frac{1}{2}\right)^n &= 2 \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} \left(\frac{1}{2}\right)^{n+1} = 2 \ln\left(\frac{3}{2}\right) = \ln\left(\frac{9}{4}\right) \text{이다.} \end{aligned}$$

18.

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad e^x - 1 = \sum_{n=1}^{\infty} \frac{x^n}{n!} \text{이므로} \\ \sum_{n=1}^{\infty} (-1)^n \frac{(\sqrt{\ln 2})^{2n-1}}{n!} \\ &= \frac{1}{\sqrt{\ln 2}} \sum_{n=1}^{\infty} (-1)^n \frac{(\sqrt{\ln 2})^{2n}}{n!} \\ &= \frac{1}{\sqrt{\ln 2}} \sum_{n=1}^{\infty} (-1)^n \frac{(\ln 2)^n}{n!} \\ &= \frac{1}{\sqrt{\ln 2}} \sum_{n=1}^{\infty} \frac{(-\ln 2)^n}{n!} \\ &= \frac{1}{\sqrt{\ln 2}} (e^{-\ln 2} - 1) \\ &= -\frac{1}{2\sqrt{\ln 2}} \end{aligned}$$

19.

$$\tan^{-1}x = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} \text{이므로}$$

$$\begin{aligned} & \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \frac{1}{3^n} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \left(\frac{1}{3}\right)^n \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \left(\frac{1}{\sqrt{3}}\right)^{2n} \\ &= \sqrt{3} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \left(\frac{1}{\sqrt{3}}\right)^{2n+1} \\ &= \frac{\sqrt{3}\pi}{6} \text{이다.} \end{aligned}$$

정리.

$1) \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$	$2) \frac{x}{(1-x)^2} = \sum_{n=1}^{\infty} nx^n$
$3) \frac{x^2+x}{(1-x)^3} = \sum_{n=1}^{\infty} n^2 x^n$	$4) \frac{x^3+4x^2+x}{(1-x)^4} = \sum_{n=1}^{\infty} n^3 x^n$

20.

$$\frac{x}{(1-x)^2} = \sum_{n=1}^{\infty} nx^n \text{이므로}$$

$$\sum_{n=1}^{\infty} \frac{n2^{n-1}}{3^n} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{n2^n}{3^n} = \frac{1}{2} \sum_{n=1}^{\infty} n \left(\frac{2}{3}\right)^n = \frac{1}{2} \frac{\frac{2}{3}}{\left(\frac{1}{3}\right)^2} = 3 \text{이다.}$$

21.

$$\begin{aligned} \ln(1+x) &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n \text{을 적분하면} \\ (1+x)\ln(1+x) - (1+x) + C &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n(n+1)} x^{n+1} \text{이다.} \\ \text{양변에 } x=0 \text{을 대입하면 } C &= 1 \text{이므로} \\ (1+x)\ln(1+x) - x &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n(n+1)} x^{n+1} \text{이다.} \\ \text{따라서 } \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n(n+1)} &= 2\ln 2 - 1 \text{이다.} \end{aligned}$$

22.

$$\begin{aligned} & \sum_{n=1}^{\infty} \frac{2}{n(n+1)(n+2)} \\ &= \sum_{n=1}^{\infty} \frac{(n+2) - n}{n(n+1)(n+2)} \\ &= \sum_{n=1}^{\infty} \left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right) \\ &= \sum_{n=1}^{\infty} \frac{1}{n(n+1)} - \sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)} \\ &= \left(\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots \right) - \left(\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots \right) \\ &= \frac{1}{2} \end{aligned}$$

23.

$$\begin{aligned} & \sum_{n=2}^{\infty} \frac{n+1}{3^n(n-1)} \\ &= \sum_{n=2}^{\infty} \frac{n+1}{n-1} \left(\frac{1}{3}\right)^n \\ &= \sum_{n=1}^{\infty} \frac{n+2}{n} \left(\frac{1}{3}\right)^{n+1} \\ &= \frac{1}{3} \sum_{n=1}^{\infty} \frac{n+2}{n} \left(\frac{1}{3}\right)^n \\ &= \frac{1}{3} \sum_{n=1}^{\infty} \left(1 + \frac{2}{n}\right) \left(\frac{1}{3}\right)^n \end{aligned}$$

$$\begin{aligned} &= \frac{1}{3} \left(\sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n + \sum_{n=1}^{\infty} \frac{2}{n} \left(\frac{1}{3}\right)^n \right) \\ &= \frac{1}{3} \left(\frac{\frac{1}{3}}{1 - \frac{1}{3}} - 2\ln\left(1 - \frac{1}{3}\right) \right) \\ &= \frac{1}{3} \left(\frac{1}{2} - 2\ln\left(\frac{2}{3}\right) \right) \\ &= \frac{1}{6} + \frac{2}{3} \ln\left(\frac{3}{2}\right) \end{aligned}$$

24.

$$\begin{aligned} \tanh^{-1}x &= \sum_{n=0}^{\infty} \frac{1}{(2n+1)} x^{2n+1} = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) \text{이므로} \\ \sum_{n=0}^{\infty} \frac{1}{2n+1} \left(\frac{1}{2}\right)^{2n} &= 2 \sum_{n=0}^{\infty} \frac{1}{2n+1} \left(\frac{1}{2}\right)^{2n+1} \\ &= 2 \tanh^{-1}\left(\frac{1}{2}\right) \\ &= \ln\left(\frac{1 + \frac{1}{2}}{1 - \frac{1}{2}}\right) \\ &= \ln(3) \end{aligned}$$

25.

$$\begin{aligned} & \sum_{n=1}^{\infty} \frac{n(n+1)}{2^n} \\ &= \sum_{n=1}^{\infty} (n^2 + n) \left(\frac{1}{2}\right)^n \\ &= \sum_{n=1}^{\infty} n^2 \left(\frac{1}{2}\right)^n + \sum_{n=1}^{\infty} n \left(\frac{1}{2}\right)^n \\ &= \left(\frac{1}{2}\right)^2 + \frac{1}{2} + \frac{1}{\left(1 - \frac{1}{2}\right)^3} + \frac{1}{\left(1 - \frac{1}{2}\right)^2} = 8 \end{aligned}$$

26.

$$\begin{aligned} & \sum_{n=3}^{\infty} \frac{(n+1)^2}{2^n(n-2)} \\ &= \sum_{n=3}^{\infty} \frac{(n+1)^2}{(n-2)} \left(\frac{1}{2}\right)^n \\ &= \sum_{n=1}^{\infty} \frac{(n+3)^2}{n} \left(\frac{1}{2}\right)^{n+2} \\ &= \frac{1}{4} \sum_{n=1}^{\infty} \left(n + 6 + \frac{9}{n}\right) \left(\frac{1}{2}\right)^n \\ &= \frac{1}{4} \left(\sum_{n=1}^{\infty} n \left(\frac{1}{2}\right)^n + \sum_{n=1}^{\infty} 6 \left(\frac{1}{2}\right)^n + \sum_{n=1}^{\infty} \frac{9}{n} \left(\frac{1}{2}\right)^n \right) \\ &= \frac{1}{4} \left(\frac{\frac{1}{2}}{\left(1 - \frac{1}{2}\right)^2} + 6 \frac{\frac{1}{2}}{1 - \frac{1}{2}} - 9\ln\left(1 - \frac{1}{2}\right) \right) \\ &= \frac{1}{4} (2 + 6 + 9\ln(2)) \\ &= 2 + \frac{9}{4} \ln(2) \end{aligned}$$

27.

$$\begin{aligned} & \sum_{n=1}^{\infty} \frac{n^2 + 3n + 1}{(n+2)!} \\ &= \sum_{n=1}^{\infty} \frac{(n+2)(n+1) - 1}{(n+2)!} \\ &= \sum_{n=1}^{\infty} \frac{1}{n!} - \sum_{n=1}^{\infty} \frac{1}{(n+2)!} \\ &= 1 + \frac{1}{2} = \frac{3}{2} \end{aligned}$$

28.

$$f(x) = \sum_{n=0}^{\infty} (n+1)x^{2n} = 1 + 2x^2 + 3x^4 + 4x^6 \dots$$

$$\Rightarrow f'(x) = \sum_{n=0}^{\infty} 2n(n+1)x^{2n-1}$$

$$\Rightarrow \frac{x}{2} f'(x) = \sum_{n=0}^{\infty} (n^2 + n)x^{2n}$$

$$\Rightarrow \frac{x}{2} f'(x) = \sum_{n=0}^{\infty} n^2 x^{2n} + \sum_{n=0}^{\infty} nx^{2n}$$

$$\Rightarrow \sum_{n=0}^{\infty} n^2 x^{2n} = \frac{x}{2} f'(x) - \sum_{n=0}^{\infty} nx^{2n}$$

$$\text{여기서 } \sum_{n=0}^{\infty} nx^{2n} = x^2 + 2x^4 + 3x^6 + \dots = x^2(1 + 2x^2 + 3x^4 \dots) \\ = x^2 f(x) \text{ 이므로}$$

$$\sum_{n=0}^{\infty} n^2 x^{2n} = \frac{x}{2} f'(x) - x^2 f(x) \text{ 이다.}$$

29.

$$f(x) = \frac{2}{(2-x)^2}$$

$$= \frac{1}{2} \frac{1}{\left(1 - \frac{x}{2}\right)^2}$$

$$= \frac{1}{2} \sum_{n=1}^{\infty} n \left(\frac{x}{2}\right)^{n-1} \left(\because \frac{x}{(1-x)^2} = \sum_{n=1}^{\infty} nx^n \right)$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} (n+1) \left(\frac{x}{2}\right)^n$$

$$= \sum_{n=0}^{\infty} \frac{(n+1)}{2^{n+1}} x^n$$

$$\therefore a_n = \frac{(n+1)}{2^{n+1}}$$

$$f'''(x) = \sum_{n=0}^{\infty} \frac{(n+1)}{2^{n+1}} n(n-1)(n-2)x^{n-3}$$

$$\Rightarrow x f'''(x) = \sum_{n=0}^{\infty} \frac{(n+1)}{2^{n+1}} n(n-1)(n-2)x^{n-2}$$

$$\Rightarrow x f'''(x) = \sum_{n=3}^{\infty} \frac{(n+1)}{2^{n+1}} n(n-1)(n-2)x^{n-2}$$

$$\Rightarrow x f'''(x) = \sum_{n=1}^{\infty} \frac{(n+3)}{2^{n+3}} (n+2)(n+1)nx^n$$

$$\therefore b_n = \frac{n+3}{2^{n+3}} (n+2)(n+1)(n)$$

$$\lim_{n \rightarrow \infty} \frac{n^3 a_n}{b_n} = 4$$

30.

$$f(x) = x^{\frac{1}{2}}, f'(x) = \frac{1}{2} x^{-\frac{1}{2}}, f''(x) = -\frac{1}{4} x^{-\frac{3}{2}}, f'''(x) = \frac{3}{8} x^{-\frac{5}{2}} \text{ 이므로}$$

$$\text{구간 } [3,5] \text{에서 } |f'''(x)| \leq \frac{3}{8} \frac{1}{3^{\frac{5}{2}}} \text{ 이다. 또한, } |x-4| \leq 1 \text{ 이므로}$$

테일러 부등식에 의한 오차의 한계는

$$\frac{3}{8} \frac{1}{3^{\frac{5}{2}}} \\ |R_2(x)| \leq \frac{3^{\frac{3}{2}}}{3!} 1^3 = 0.004 \text{ 이다. } \left(\because \frac{1}{3^{\frac{5}{2}}} \approx 0.064 \right)$$